AD HOC INSIGHTS

Consumer Goods

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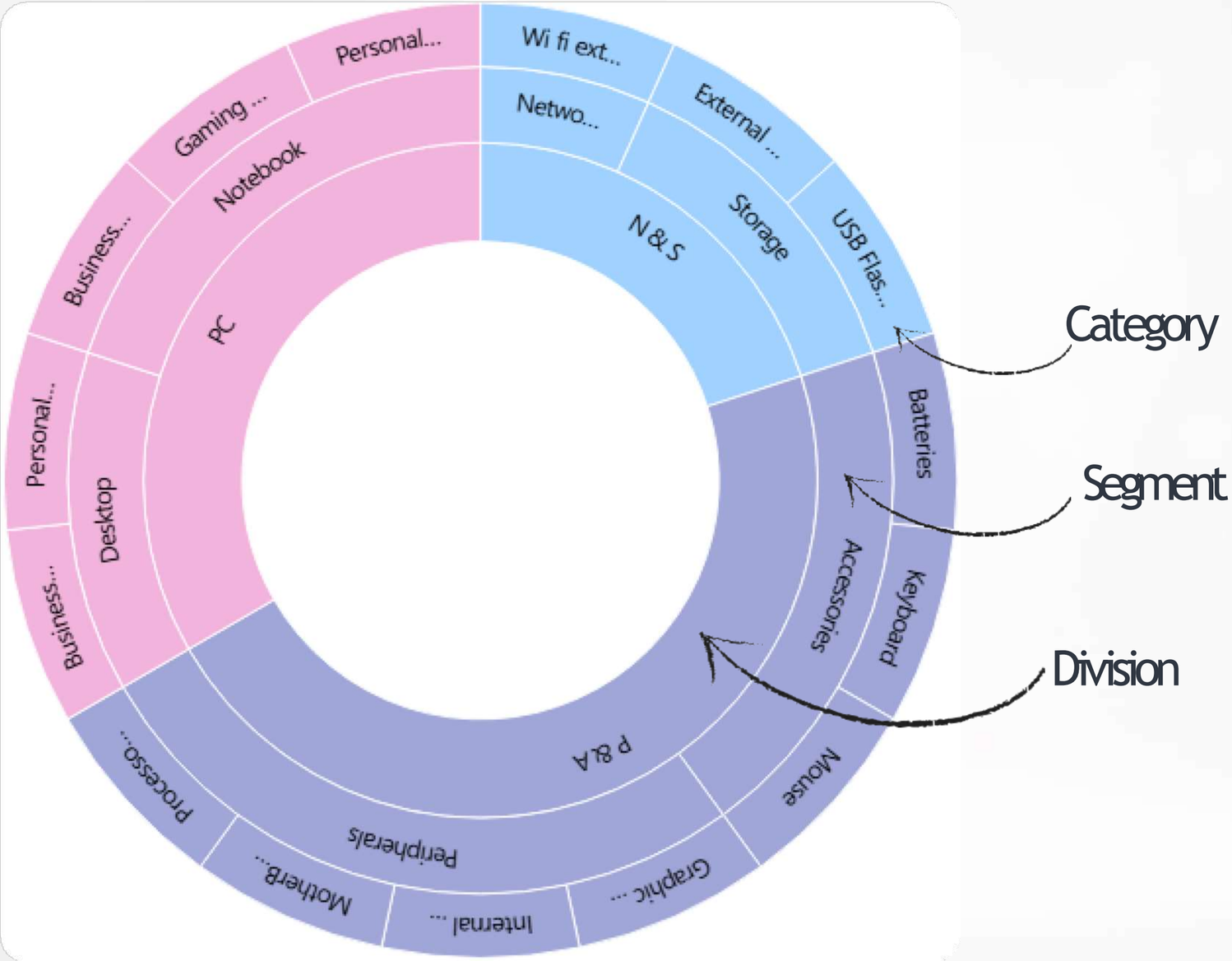
OBJECTIVES

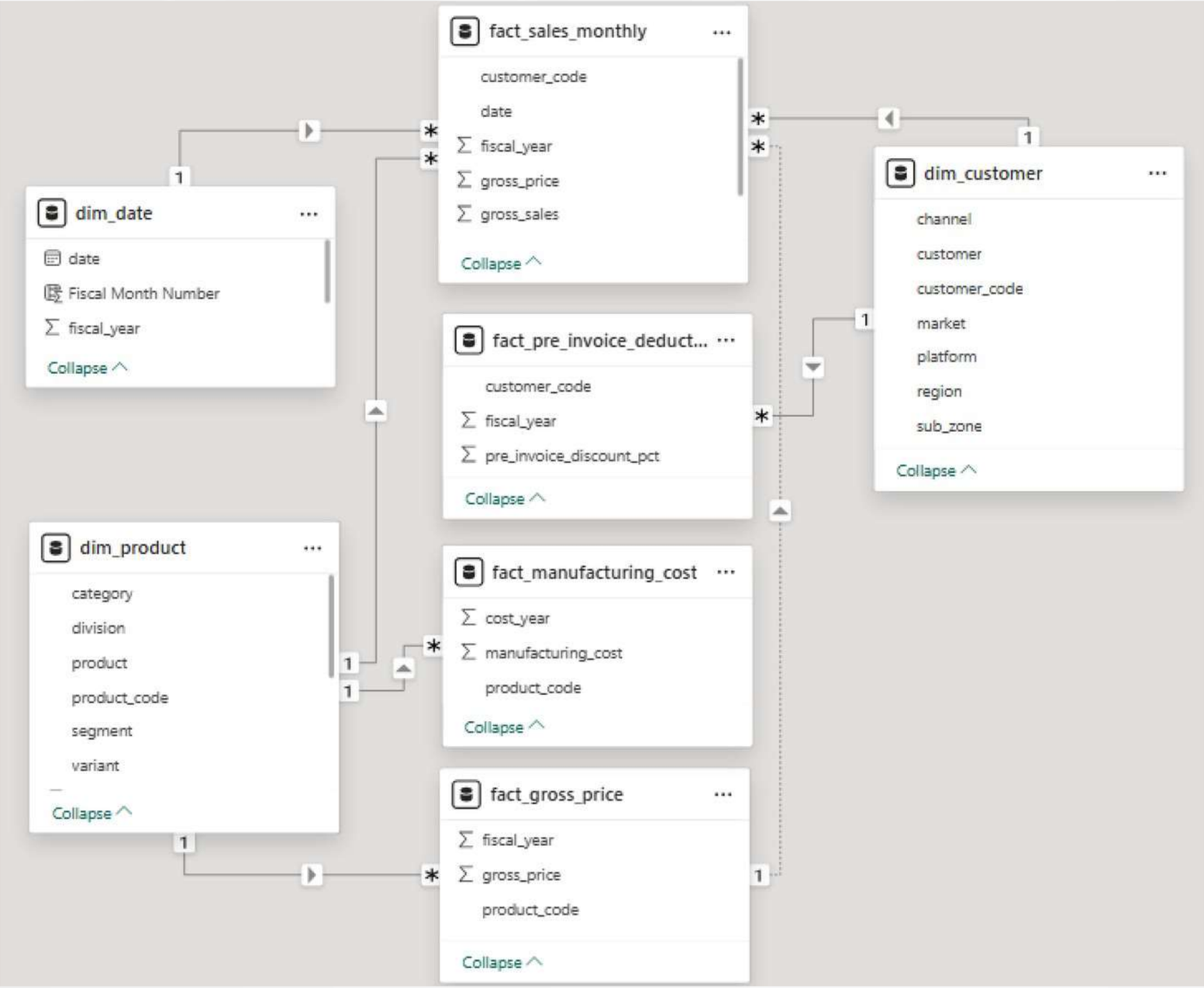
- **AtliQ Hardware** is a fictitious company operating in the computer hardware industry with presence across multiple markets
- The management **lacks sufficient insights** to make quick and data-informed decisions
- The objective of this project is to **analyze business data** and generate meaningful insights
- Solve 10 ad-hoc business requests using SQL
- Build visualizations in Power BI to present insights in a clear and understandable way



COMPANY DETAILS

- AtliQ Hardware's **fiscal year** starts from **September to August**.
- AtliQ Hardware is a manufacturer of **computer hardware** across various different **divisions, segments,** and **categories.**

MARKET PRESENCE

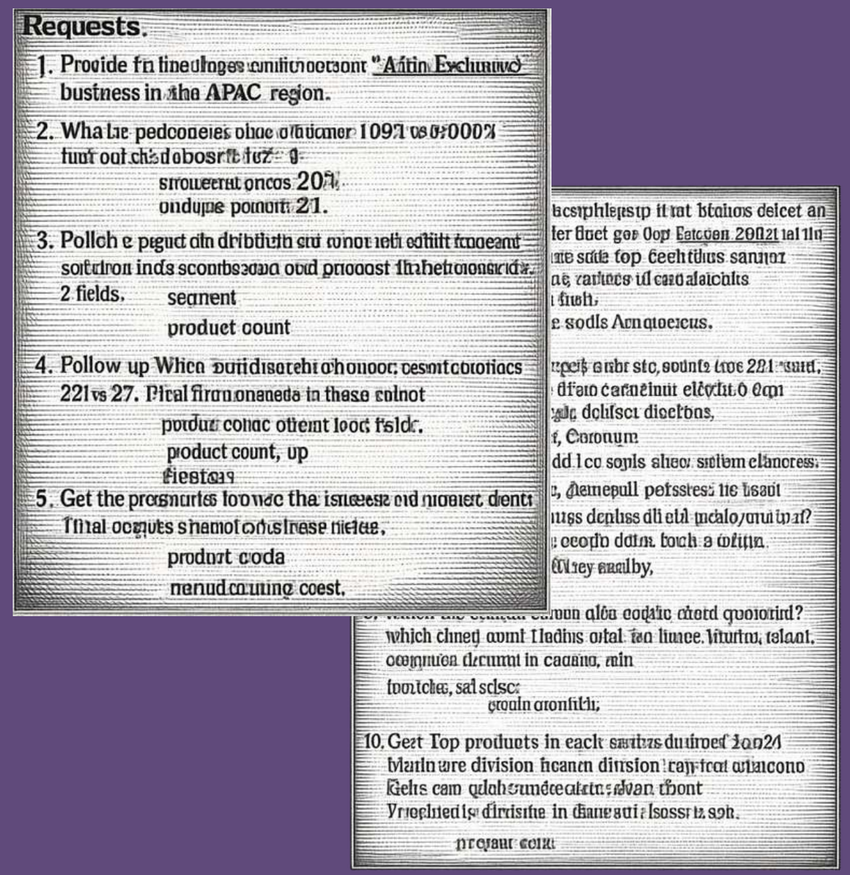


DATA MODEL

The data model consisted of a **star schema** with **4 fact tables** and **3 dimension tables.**

BUSINESS QUESTIONS (AD HOC REQUESTS)

- 10 business questions were provided to derive insights
- Each request was solved using SQL queries
- Results were visualized using Power BI dashboards



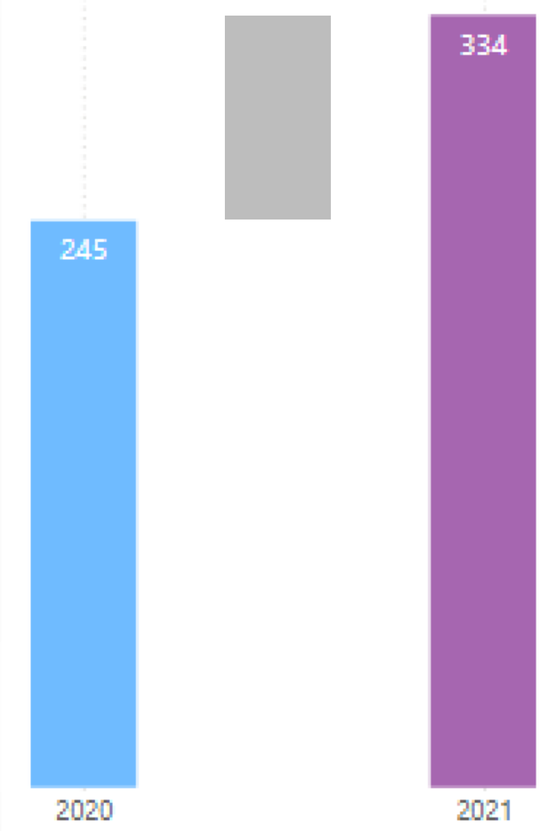
1. Provide the list of markets in which customer Atliq Exclusive operates its business in the APAC region.

market
India
Indonesia
Japan
Philiphines
South Korea
Australia
Newzealand
Bangladesh

2. What is the percentage of unique product increase in 2021 vs. 2020?

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33

36.33%
Percentage
Increase



Unique products 2020

Unique products 2021

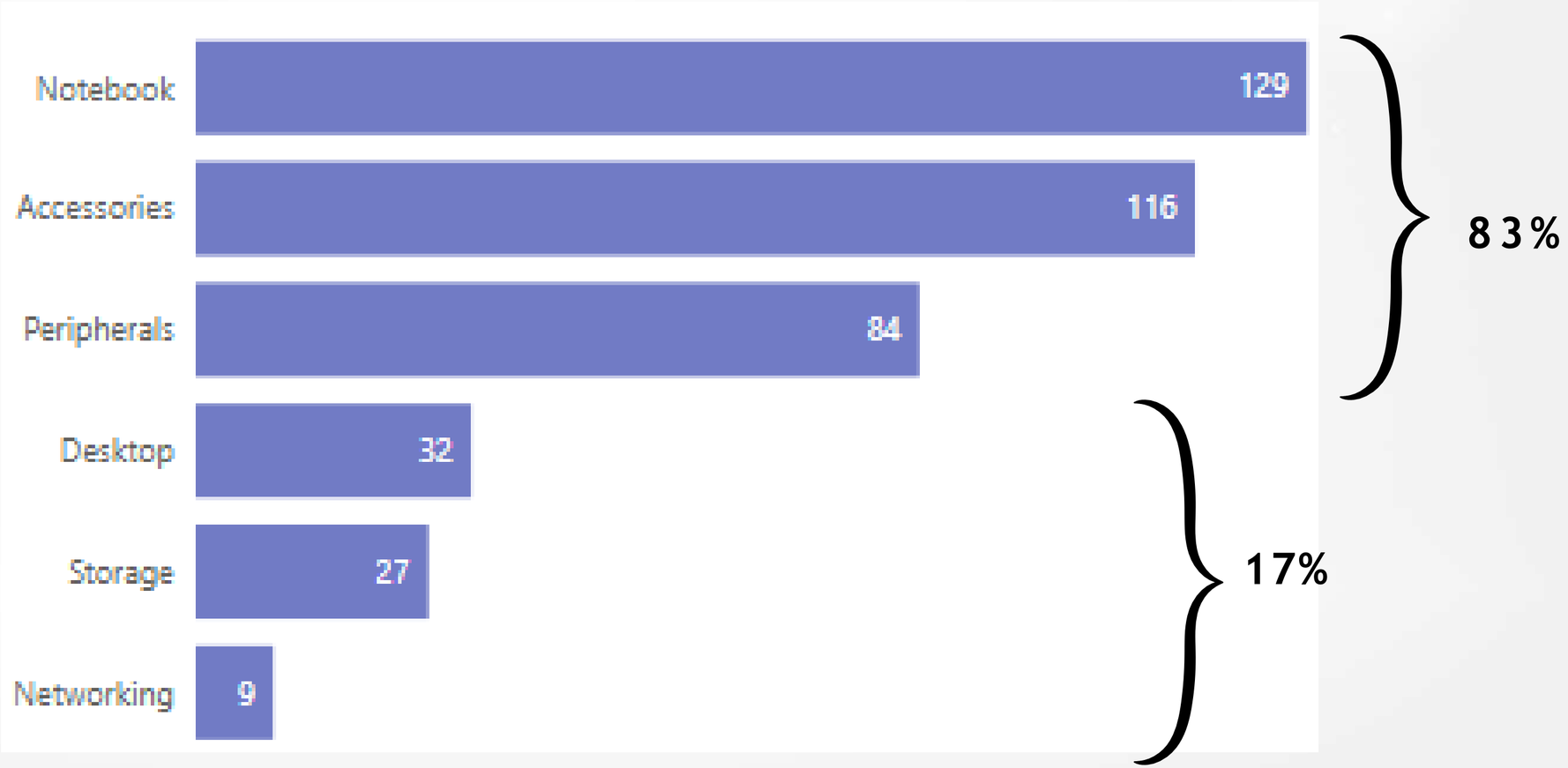
Insight:

Production and demand both increased significantly.

3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

Unique **product counts** for each **segment**



Insight:

- Notebooks, Accessories and Peripherals show **accelerated manufacturing growth** constituting to ~83% of the total unique product growth
- Further analysis using sales or revenue metrics is required to determine whether lower-product segments are actually underperforming.

4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

segment	products_count_2020	products_count_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

segment	Products 2020	Products 2021	Difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

Insight:

Accessories segment is dominating in terms of highest difference in **positive growth.**

Storage and Networking segment's production **growth is slower** compared to other segments.

5. Get the products that have the highest and lowest manufacturing costs.

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Insight:

A **wide gap** in manufacturing cost indicates **varying product complexity**, requiring category-specific pricing and profitability strategies.

Highest Manufacting Cost

A6120110206
AQ HOME Allin1 Gen 2

$240.54

Personal Desktop

Lowest Manufacting Cost

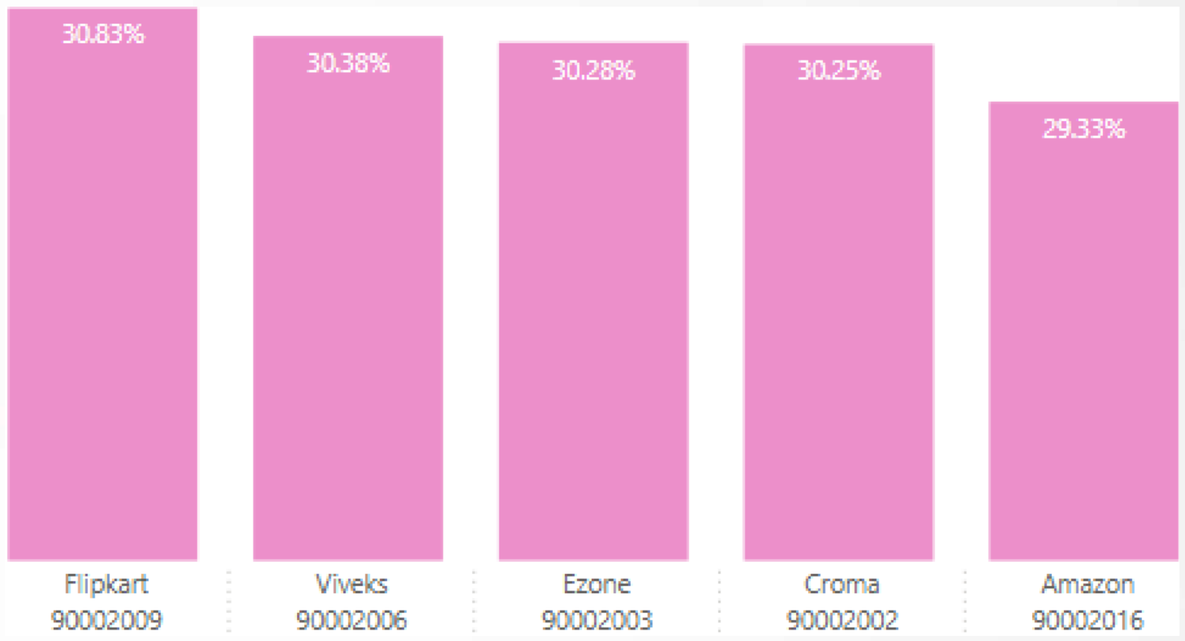
A2118150101
AQ Master wired x1 Ms

$0.89

Mouse

6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

customer_code	customer	average_discount_precentage
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933



Customer and **Customer Code**

Insight:

- The top customers in the Indian market receive **consistently high discounts (~30%)**, indicating a standardized **discounting strategy** for key accounts.
- These customers are likely **major partners** or **high-volume buyers**, justifying higher negotiated discounts.

7. Get the complete report of the Gross sales amount for the customer Atliq Exclusive for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

date	fiscal_year	gross_sales
2019-09-01	2020	4496259.67
2019-10-01	2020	5135902.35
2019-11-01	2020	7522892.56
2019-12-01	2020	4830404.73
2020-01-01	2020	4740600.16
2020-02-01	2020	3996227.77
2020-03-01	2020	378770.97
2020-04-01	2020	395035.35
2020-05-01	2020	783813.42
2020-06-01	2020	1695216.60
2020-07-01	2020	2551159.16
2020-08-01	2020	2786648.26

FY 2020

E 39.31 M

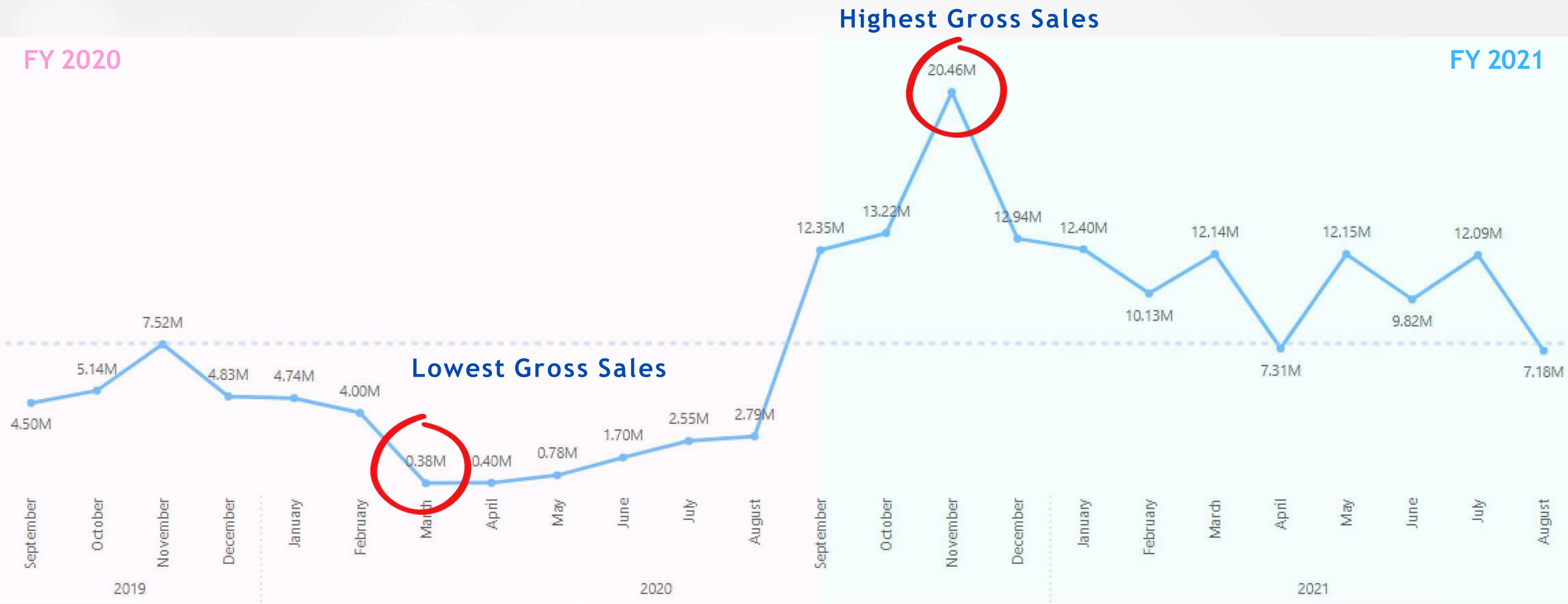
2020-09-01	2021	12353509.79
2020-10-01	2021	13218636.20
2020-11-01	2021	20464999.10
2020-12-01	2021	12944659.65
2021-01-01	2021	12399392.98
2021-02-01	2021	10129735.57
2021-03-01	2021	12144061.25
2021-04-01	2021	7311999.95
2021-05-01	2021	12150225.01
2021-06-01	2021	9824521.01
2021-07-01	2021	12092346.32
2021-08-01	2021	7178707.59

FY 2021

E 142.21 M

Insight:

- Gross sales show a sharp decline around March–April 2020, followed by strong recovery and growth in FY 2021.
- FY 2021 consistently outperforms FY 2020, contributing to about **78%** of total gross sales, indicating improved business performance after the initial dip.



COVID-19 disruptions in early 2020 may have impacted sales temporarily.
Source: Link

Laptops, Cell Phones, eGamesDefied Slump as COVID-19 Dented 2020's Electronics Sale

News / By ewastemonitor

Foreseen drop in 2020 sales-related e-waste : 4.9 million metric tonnes or ~6.4%;
Pandemic intensified digital divide between high, middle and low income countries

Lockdown: India sees sharp dip in office hardware peripherals demand, says IDC

8. In which quarter of 2020, got the maximum total_sold_quantity?

qtr	total_sold_quantity
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087

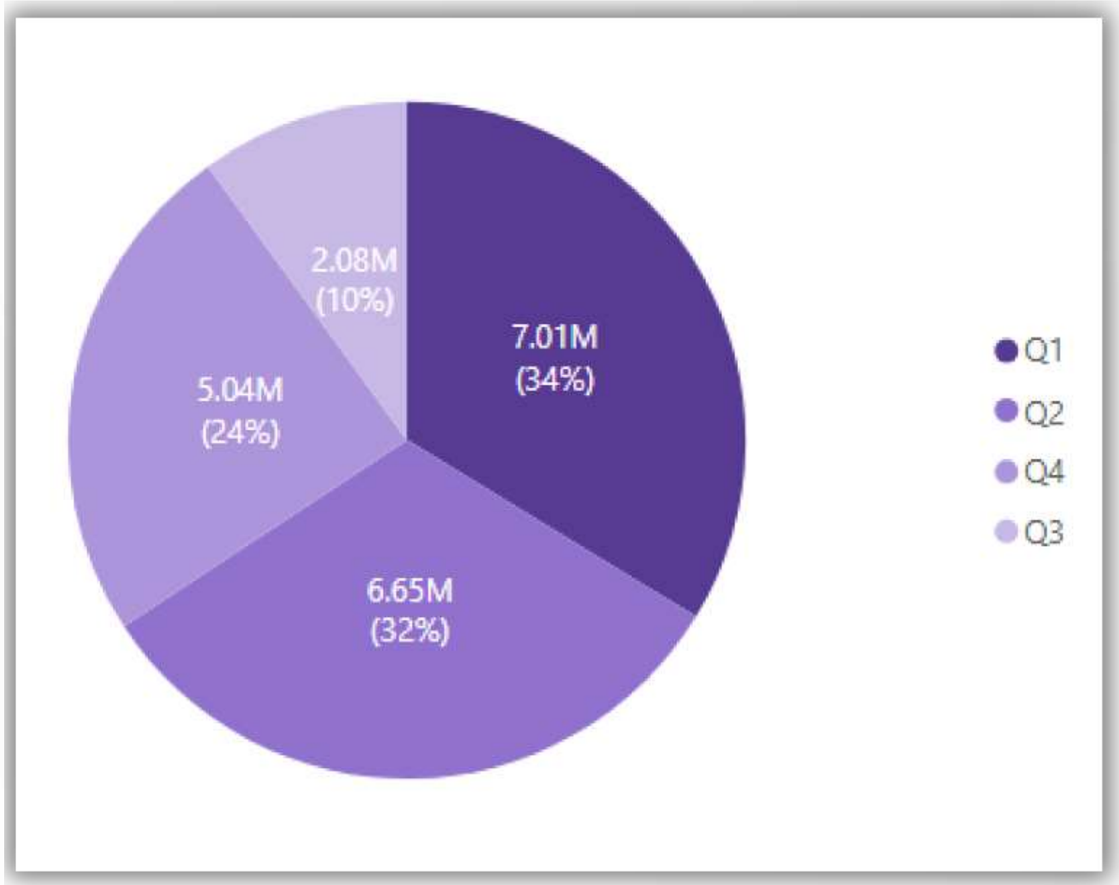
Quarter	Total Sold Quantity
Q1	**7.01M**
September	1.76M
October	2.19M
November	3.05M
Q2	**6.65M**
January	1.76M
February	1.70M
December	3.18M
Q3	**2.08M**
March	0.24M
April	0.82M
May	1.02M
Q4	**5.04M**
June	1.56M
July	1.69M
August	1.79M

Insight:

- **Q1** contributed the **highest share** (~**34%**) of total sold quantity in FY 2020, followed closely by Q2.
- **Q3** recorded the **lowest contribution** (~**10%**), indicating a significant dip in demand during this period.
- This uneven distribution suggests **strong seasonality**, with certain quarters consistently outperforming others.

Total Sold Quantity in **FY 2020** by **Quarter**

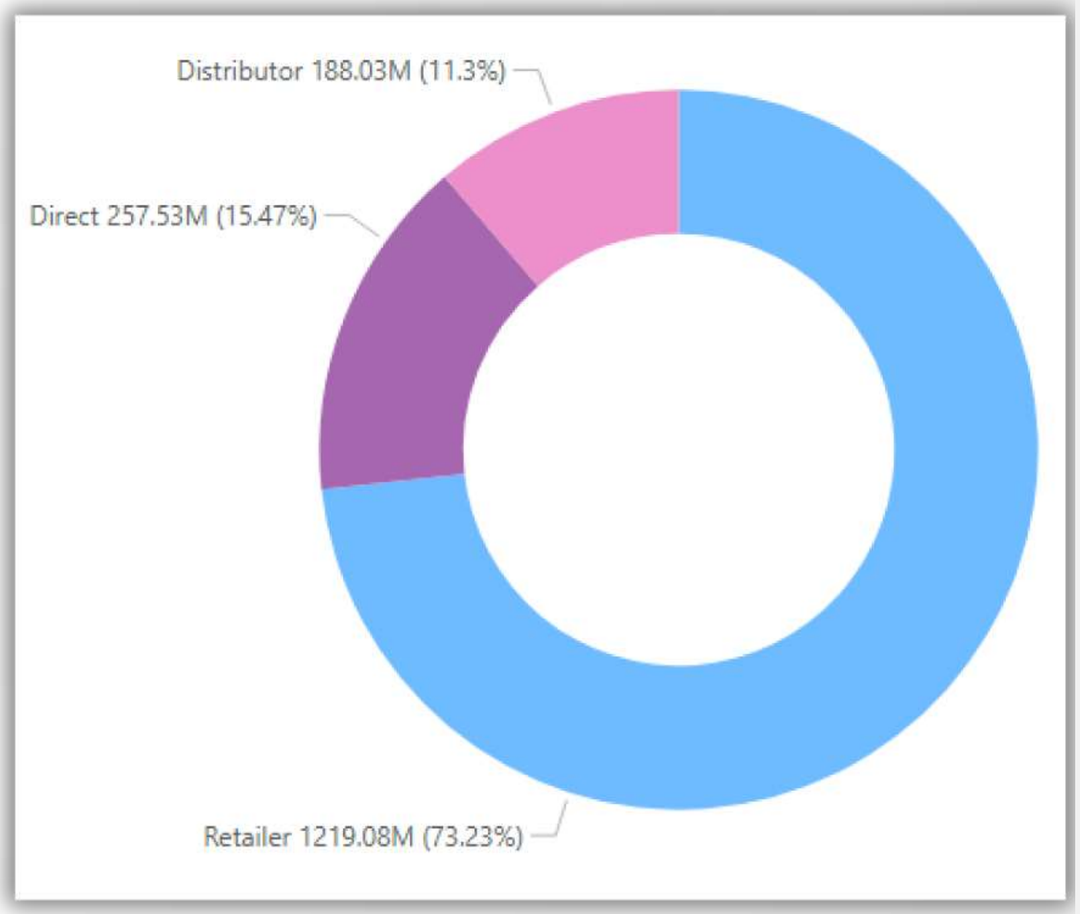
Month	Quarter	Total Sold Quantity
September	Q1	1.76M
October	Q1	2.19M
November	Q1	3.05M
December	Q2	3.18M
January	Q2	1.76M
February	Q2	1.70M
March	Q3	0.24M
April	Q3	0.82M
May	Q3	1.02M
June	Q4	1.56M
July	Q4	1.69M
August	Q4	1.79M



9. Which channel helped to bring more gross sales in the fiscal year 2021 and what is its percentage of contribution?

channel	gross_sales_mln	percentage
Retailer	1219.08M	73.23%
Direct	257.53M	15.47%
Distributor	188.03M	11.30%

Gross Sales and % Contribution by **Channel** for **FY 2021**



Insights:

The **Retailer** channel contributes the majority of gross sales (~**73%**) in FY 2021, making it the **primary revenue driver.**

Direct and Distributor channels contribute significantly less, indicating a heavy reliance on retail partners for sales.

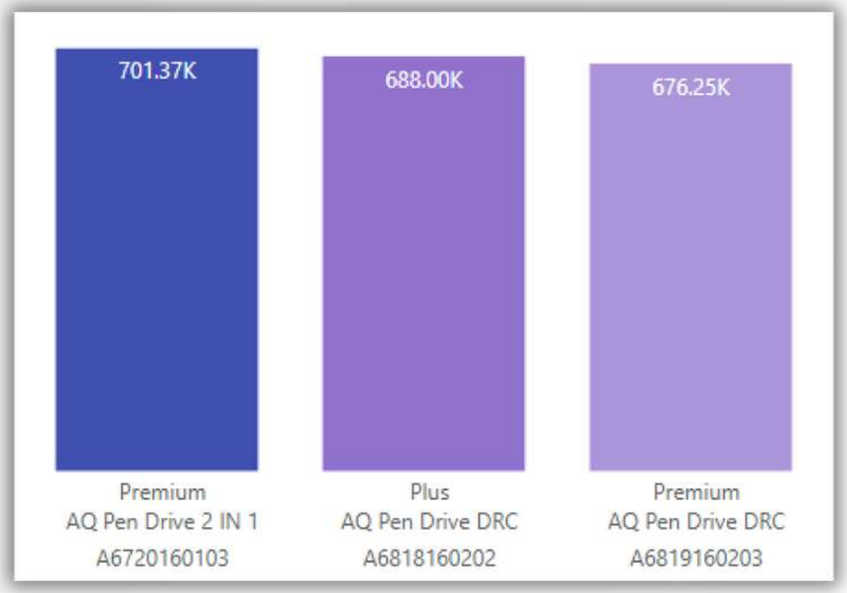
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1 [Premium]	701373	1
N & S	A6818160202	AQ Pen Drive DRC [Plus]	688003	2
N & S	A6819160203	AQ Pen Drive DRC [Premium]	676245	3
P & A	A2319150302	AQ Gamers Ms [Standard 2]	428498	1
P & A	A2520150501	AQ Maxima Ms [Standard 1]	419865	2
P & A	A2520150504	AQ Maxima Ms [Plus 2]	419471	3
PC	A4218110202	AQ Digit [Standard Blue]	17434	1
PC	A4319110306	AQ Velocity [Plus Red]	17280	2
PC	A4218110208	AQ Digit [Premium Misty Green]	17275	3

Insights:

- The Retailer channel contributes the majority of gross sales (~**73%**) in FY 2021, making it the primary revenue driver.
- **Direct** and **Distributor** channels contribute **significantly less**, indicating a heavy reliance on retail partners for sales.

Top 3 Products sold in each **division** by **gross sales** in **FY 2021**



Divison: **N & S**

Divison: **P & A**

Divison: **PC**

BUSINESS RECOMMENDATIONS

1. Reduce dependency on Retail channel by strengthening Direct and Distributor channels to improve channel diversification.

2. Leverage high-performing months (e.g., November) with targeted campaigns and inventory planning to maximize revenue.

3. Implement strategies to boost sales during low-performing periods (e.g., Q3), such as discounts, promotions, or bundled offers.

4. Focus on expanding successful product lines by introducing more variants of top-performing products.

5. Continuously evaluate discount strategies to balance customer retention and profitability.

THANK YOU

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